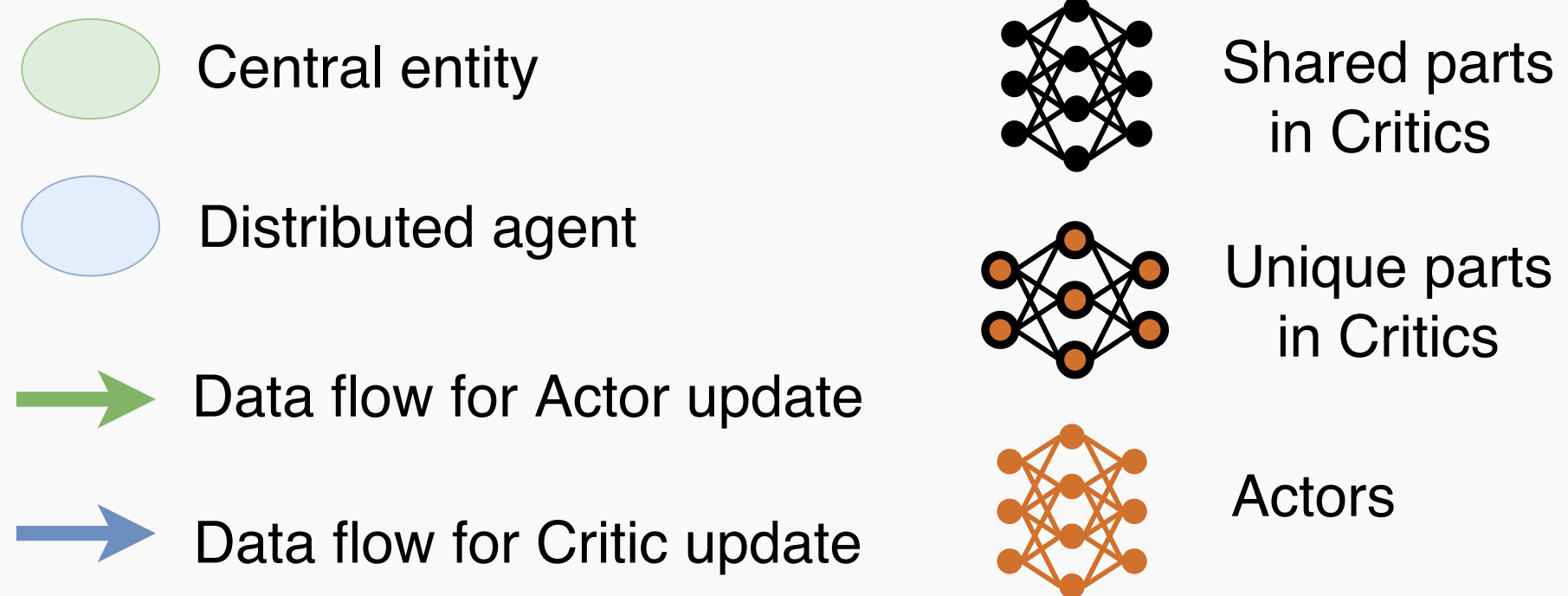
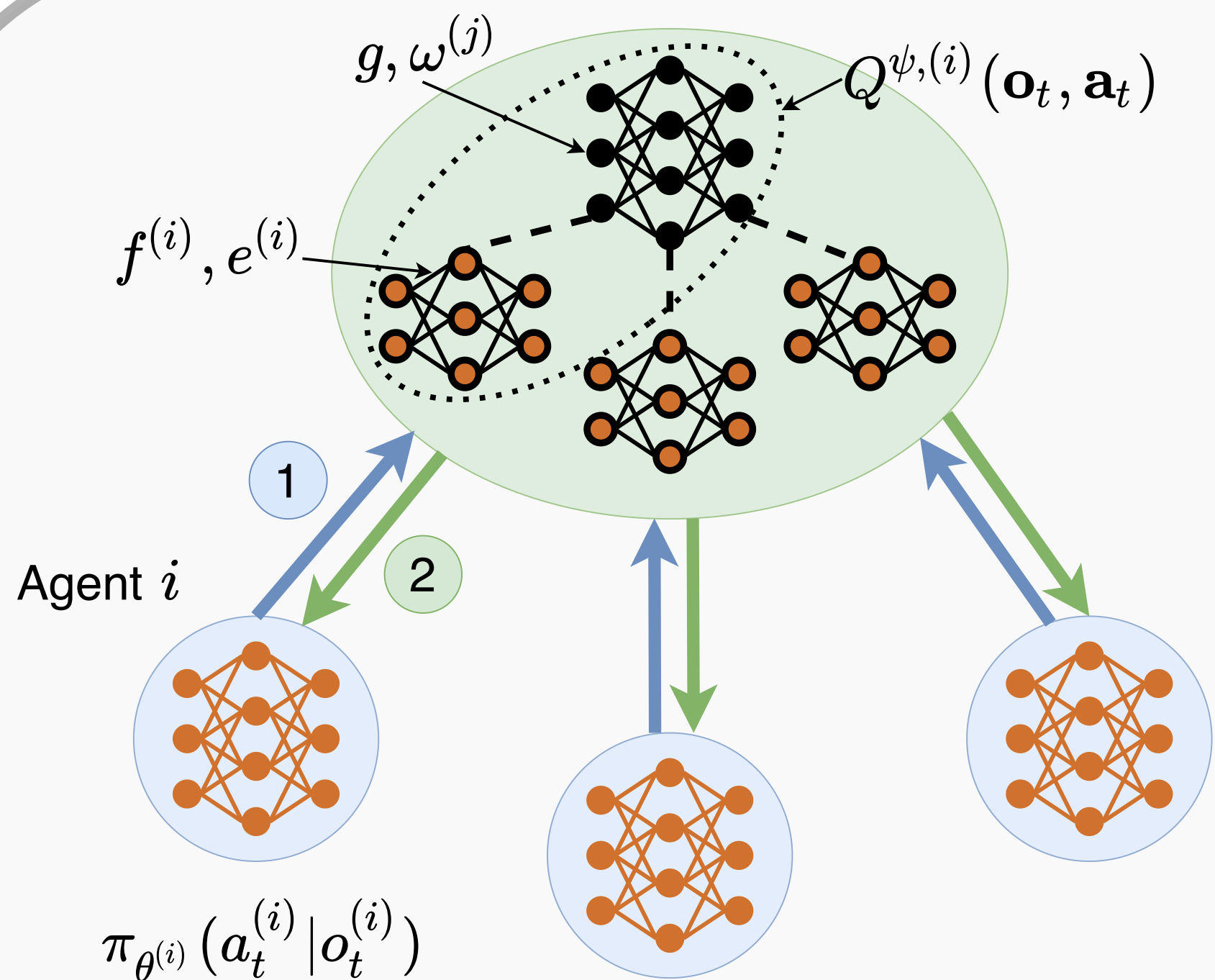
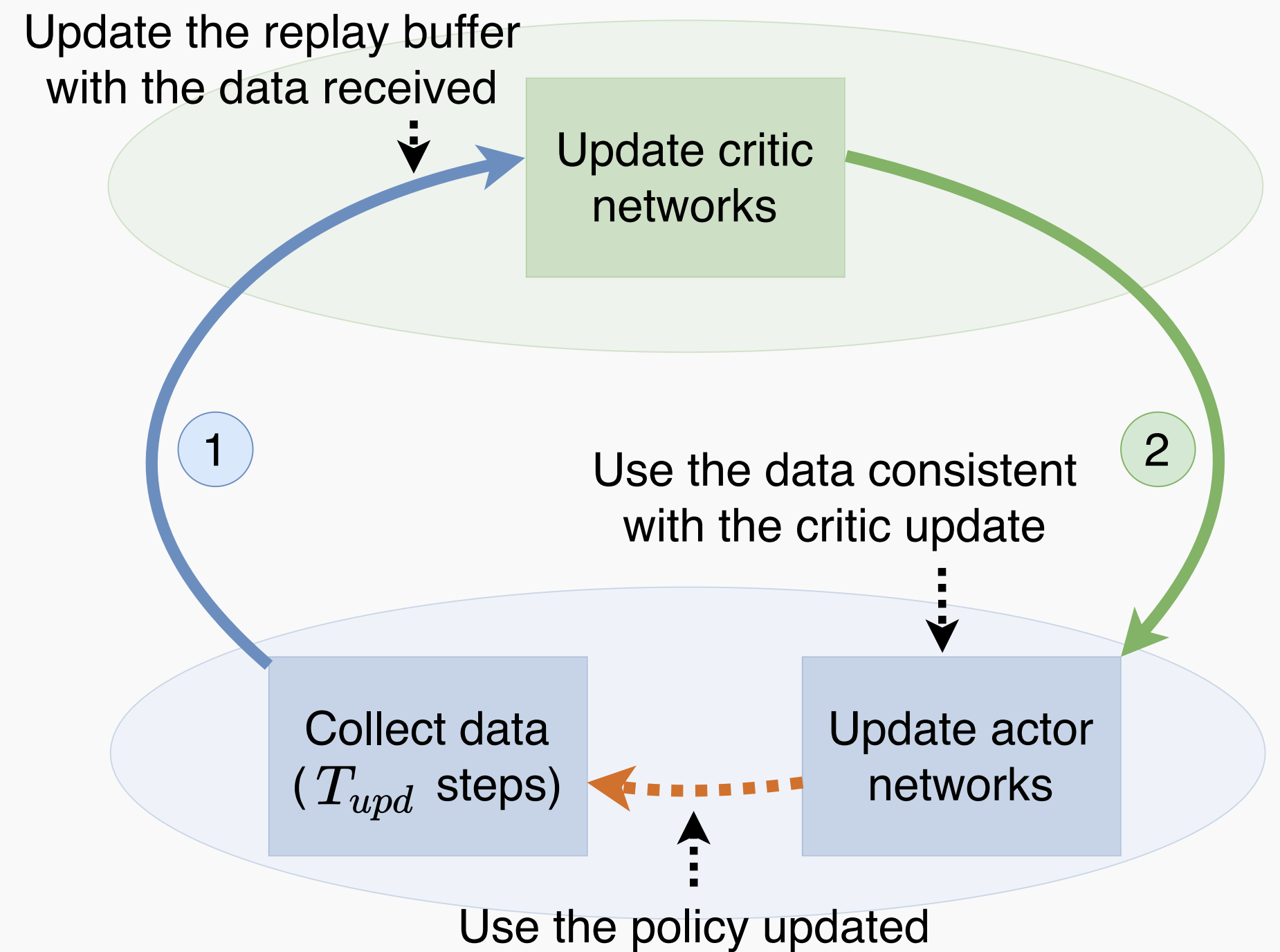




(a) Schematic illustration of the model architecture



1

$$\{(o_t^{(i)}, a_t^{(i)}, r_t^{(i)}, o_{t+1}^{(i)})\}_{T_{upd}}$$

$$\bar{a}_{t+1}^{(i)} \sim \pi_{\bar{\theta}^{(i)}}(\bar{a}_{t+1}^{(i)} | o_{t+1}^{(i)})$$

$$\log(\pi_{\bar{\theta}^{(i)}}(\bar{a}_{t+1}^{(i)} | o_{t+1}^{(i)}))$$

$$\hat{a}_t^{(i)} \sim \pi_{\theta^{(i)}}(\hat{a}_t^{(i)} | o_t^{(i)})$$

2

$$\{o_t^{(i)}\}_{h_{size}} \subset h$$

$$Q^{\psi, (i)}(o_t, \hat{a}_t)$$

$$b(o_t, \hat{a}_t^{(\setminus i)})$$

(b) Information exchanges and model updates